

Theoretical Separation of Quantum Vacuum Energy

from the Cosmological Constant: A Review of Foundational Approaches

Boris Kriger^{1,2}

¹ Information Physics Institute, Gosport
`boris.kriger@informationphysicsinstitute.net`

² Institute of Integrative and Interdisciplinary Research, Toronto
`boriskriger@interdisciplinary-institute.org`

ORCID: [0009-0001-0034-2903](https://orcid.org/0009-0001-0034-2903)

Abstract

The identification of the quantum vacuum energy density ρ_{vac} with the cosmological constant Λ , proposed by Zel'dovich in 1967, has produced the most severe quantitative discrepancy in the history of physics—a mismatch of 55–120 orders of magnitude between theoretical estimates and observational bounds. This paper reviews and synthesizes multiple independent theoretical frameworks in which Λ and ρ_{vac} are treated as physically distinct entities. We examine geometric formulations of general relativity in which Λ emerges as an integration constant (trace-free Einstein gravity, unimodular gravity, Codazzi-equation approaches); thermodynamic and condensed-matter analogies showing that vacuum energy is governed by macroscopic boundary conditions rather than zero-point fluctuations; renormalization-group methods demonstrating that ρ_{vac} runs dynamically with the Hubble rate H ; and recent effective-field-theory constructions—in particular the Quantum-Kinetic Dark Energy (QKDE) framework—in which a time-dependent scalar kinetic normalization carries the dark-energy phenomenology while the Einstein–Hilbert metric sector and the Planck mass remain unmodified. We develop the mathematical structure underlying each approach, prove key propositions establishing the logical independence of Λ and ρ_{vac} , and identify falsifiable observational predictions. The common thread is that the observed late-time acceleration need not equal the quantum vacuum energy density, potentially reframing—if not resolving—the cosmological constant problem.

Keywords: cosmological constant, vacuum energy, quantum field theory, general relativity, effective field theory of dark energy, running vacuum, unimodular gravity, QKDE, Hubble tension

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1. Introduction

The cosmological constant Λ was introduced by Einstein in 1917 as a geometric modification of his field equations,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (1)$$

where $G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu}$ is the Einstein tensor. Einstein’s original intent was to permit a static universe; once the expansion was established observationally by Hubble, the term was widely regarded as superfluous for decades.

A qualitative shift occurred when Zel’dovich [1, 2] observed that the stress-energy tensor of the quantum vacuum takes the Lorentz-invariant form $\langle T_{\mu\nu} \rangle_{\text{vac}} = -\rho_{\text{vac}} g_{\mu\nu}$, which is mathematically identical to a cosmological-constant contribution with $\Lambda = 8\pi G \rho_{\text{vac}}$. This identification elevated Λ from a geometric free parameter to a prediction of quantum field theory (QFT).

The prediction, however, is spectacularly wrong. Depending on the regularization scheme and the assumed ultraviolet cutoff, QFT estimates of ρ_{vac} exceed the observationally inferred value $\Lambda_{\text{obs}} \sim 10^{-47} \text{ GeV}^4$ by 55–120 orders of magnitude [3, 4, 5]. This discrepancy is commonly referred to as the *cosmological constant problem* and has been called “the worst prediction in the history of physics.”

The standard response treats the mismatch as evidence for an extraordinarily precise cancellation—either via fine-tuning, supersymmetry at unbroken scales, anthropic selection in a landscape [6, 7], or some unknown dynamical mechanism. However, a growing body of work suggests a more conservative possibility: *the identification itself may be unwarranted*. If Λ and ρ_{vac} are logically and physically distinct, the 55–120 order discrepancy is not a problem requiring exotic cancellation but rather evidence that two different quantities have been conflated.

This review synthesizes the principal approaches to such a separation. We examine:

- (i) Geometric formulations in which Λ arises as an integration constant (Section 5).
- (ii) Thermodynamic and condensed-matter analogies (Section 6).
- (iii) Running vacuum models from QFT renormalization (Section 7).
- (iv) Effective-field-theory frameworks, specifically the Quantum-Kinetic Dark Energy (QKDE) framework recently developed by Brown [13] (Section 8).
- (v) Conceptual and foundational analyses of the assumptions underlying the Zel’dovich identification (Section 9).

We develop the mathematical content of each approach with enough rigor to make the logical structure transparent and to identify falsifiable predictions.

2. Logical Structure of the Argument

The overall reasoning proceeds through five largely independent lines of evidence, each supporting the same conclusion: the identification $\Lambda = 8\pi G \rho_{\text{vac}}$ is not a necessary consequence of established physics. The logical architecture is as follows.

Step 1 (Geometric independence). In several well-defined reformulations of general relativity—including trace-free (unimodular) gravity and Codazzi-equation formulations—the cosmological constant arises as an integration constant whose value is fixed by boundary conditions, not by the vacuum stress tensor. This establishes that Λ can be a geometric quantity independent of ρ_{vac} .

Step 2 (Thermodynamic sequestration). Condensed-matter analogies, principally Volovik’s analysis of quantum liquids, demonstrate that the ground-state energy of a quantum many-body system need not determine the macroscopic “cosmological constant” of the emergent low-energy theory. The analogous quantity is instead controlled by macroscopic parameters (external pressure, system size), providing a physical mechanism by which ρ_{vac} decouples from Λ .

Step 3 (Dynamical vacuum energy). Renormalization-group computations in QFT on curved spacetime show that ρ_{vac} is not a constant but evolves as $\rho_{\text{vac}}(H) = \rho_{\text{vac}}^{(0)} + \nu H^2 + \dots$ where ν is a small dimensionless coefficient. A quantity that runs with the Hubble rate cannot be identified with a *constant* in the Einstein equations.

Step 4 (Effective-field-theory realization). Brown’s QKDE framework [13] provides an explicit construction in which all late-time dark-energy phenomenology enters through a time-dependent scalar kinetic normalization $K(\chi)$ while the Einstein–Hilbert metric sector remains exactly GR. In the Effective Field Theory of Dark Energy (EFT-DE) language, the Bellini–Sawicki parameters satisfy $\alpha_B = \alpha_M = \alpha_T = 0$ with $\alpha_K > 0$. This realizes the separation at the level of an action principle: the scalar sector carries the “vacuum-like” energy without modifying Λ or the Planck mass. Brown’s formalism—including his closed e-fold system, his algebraic identity for K'/K , and his comprehensive stability analysis—provides the mathematical machinery that elevates the separation thesis from a conceptual hypothesis to a calculable, falsifiable framework.

Step 5 (Foundational reassessment). Conceptual and historical analysis (Rugh–Zinkernagel) reveals that the Zel’dovich identification rests on interpretive choices—treating zero-point energy as gravitationally active, extrapolating the Standard Model

to Planck scales, and identifying the bare cosmological constant with the observed value—each of which can be questioned within established physics.

Synthesis. The five steps converge on a common conclusion: Λ and ρ_{vac} play distinct roles—geometric and quantum-field-theoretic, respectively—and the cosmological constant problem may be an artifact of conflating them. The remainder of the paper develops each step with rigorous mathematical content.

3. Background and Related Work

3.1. The cosmological constant problem

The modern statement of the problem begins with the semiclassical Einstein equation

$$G_{\mu\nu} + \Lambda_{\text{bare}} g_{\mu\nu} = 8\pi G (\langle T_{\mu\nu} \rangle_{\text{ren}} + T_{\mu\nu}^{(\text{matter})}), \quad (2)$$

where $\langle T_{\mu\nu} \rangle_{\text{ren}}$ is the renormalized vacuum expectation value. The observed effective cosmological constant is

$$\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + 8\pi G \rho_{\text{vac}}. \quad (3)$$

The problem is that ρ_{vac} computed from field-theoretic zero-point sums, even after Lorentz-covariant regularization, vastly exceeds $\Lambda_{\text{eff}}/8\pi G$. For a scalar field of mass m with a Planck-scale cutoff, the one-loop estimate gives $\rho_{\text{vac}} \sim M_{\text{Pl}}^4/(16\pi^2)$, while observation requires $\Lambda_{\text{eff}} \sim 10^{-122} M_{\text{Pl}}^2$ [3, 4].

3.2. Existing review literature

The cosmological constant problem has been reviewed extensively from multiple perspectives. Weinberg [3] provided the definitive early treatment. Carroll [4] and Padmanabhan [5] offered modern surveys emphasizing the field-theoretic and gravitational aspects respectively. Martin [8] analyzed the problem from the standpoint of QFT regularization. Rugh and Zinkernagel [9] addressed its philosophical and foundational dimensions. Solà Peracaula [11] has championed running-vacuum approaches, while Feng and Chen [12] systematically catalog geometric frameworks.

The present review differs from these predecessors in its focus on the *separation thesis*—the idea that Λ and ρ_{vac} can be decoupled—and in its systematic integration of Brown’s Quantum-Kinetic Dark Energy (QKDE) effective-field-theory framework [13] as an explicit, mathematically rigorous realization of that separation at the action level. Brown’s work, published in 2026 in the *International Journal of Modern Physics D*, provides a covariantly completed scalar-field construction in which the kinetic normalization $K(\chi)$ carries all dark-energy phenomenology while the Einstein–Hilbert metric

sector remains unmodified—precisely the kind of architecture required to make the separation thesis concrete and testable.

3.3. Section Result and Implications

The cosmological constant problem is a well-defined quantitative discrepancy whose resolution depends critically on whether Λ and ρ_{vac} must be identified. Existing literature treats them as coupled; the present paper systematically examines the consequences of treating them as independent.

4. Conceptual Framework

Before entering the mathematical development, we formulate the conceptual distinction at the center of this review.

Definition 4.1 (Geometric cosmological constant). The *geometric cosmological constant* Λ_{geom} is a free parameter (or integration constant) appearing in the gravitational field equations, whose value is fixed by boundary conditions or initial data and is *independent* of the matter content of the theory.

Definition 4.2 (Quantum vacuum energy density). The *quantum vacuum energy density* ρ_{vac} is the renormalized zero-point energy density of quantum fields, computed via a well-defined regularization and renormalization prescription in a given spacetime background. It is a property of the quantum matter sector.

Remark 4.3 (Conceptual claim: Non-necessity of the Zel’dovich identification). The equation $\Lambda_{\text{geom}} = 8\pi G \rho_{\text{vac}}$ is not a theorem of either general relativity or quantum field theory taken separately. It is an additional physical hypothesis whose validity depends on assumptions about the coupling between vacuum stress-energy and spacetime geometry. *Note:* This is a conceptual claim about the logical relationship between two theories, not a mathematical theorem with formal hypotheses; the “proof” below is an argument establishing the claim.

Argument. In general relativity, Λ can be introduced as a free constant in the Einstein–Hilbert action,

$$S_{\text{grav}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda), \quad (4)$$

without reference to matter fields. In QFT on a fixed background, ρ_{vac} is computed from the functional determinant of the matter-field fluctuation operator and depends on the field content and regularization scheme, but does not determine any geometric parameter of the background. The identification requires an additional semiclassical coupling prescription, namely equation (2), together with the assumption that ρ_{vac}

sources gravity with the same coupling as ordinary matter. This prescription is not derivable from either theory alone. $\square$

Counterarguments and Responses

Objection: In the semiclassical approximation, the vacuum expectation value $\langle T_{\mu\nu} \rangle$ enters the right-hand side of the Einstein equation on the same footing as classical matter. Therefore ρ_{vac} must contribute to Λ_{eff} .

Response: The semiclassical equation is itself an approximation whose domain of validity is restricted to situations where the quantum state is close to a coherent (classical) state of the gravitational field [33, 34]. Even within the semiclassical framework, the split in (3) between Λ_{bare} and ρ_{vac} is scheme-dependent: a finite renormalization can transfer any amount between the two. The physical content lies in Λ_{eff} alone, and the question of whether Λ_{bare} is “geometric” or “matter-sourced” is underdetermined by the semiclassical formalism.

4.1. Section Result and Implications

The Zel’dovich identification is a physical hypothesis, not a mathematical theorem. This opens the door to frameworks in which Λ and ρ_{vac} are governed by different physics.

5. Mathematical Development I: Geometric Approaches

5.1. Trace-free (unimodular) Einstein gravity

The standard Einstein equations (1) have trace

$$-R + 4\Lambda = 8\pi G T, \quad (5)$$

where $T = g^{\mu\nu} T_{\mu\nu}$. Trace-free Einstein gravity removes this trace, postulating only the traceless part of the field equations:

$$R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{4} T g_{\mu\nu} \right). \quad (6)$$

This system is equivalent to the full Einstein equations with an *arbitrary* cosmological constant. Indeed, applying the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ and local energy-momentum conservation $\nabla^\mu T_{\mu\nu} = 0$ to (6), one recovers (1) with Λ as an undetermined integration constant [30, 12].

Proposition 5.1 (Λ as integration constant in trace-free gravity). *Let $(M, g_{\mu\nu})$ satisfy (6) with $\nabla^\mu T_{\mu\nu} = 0$. Then there exists a constant $\Lambda \in \mathbb{R}$ such that $(M, g_{\mu\nu})$*

satisfies (1). The value of Λ is determined by initial or boundary data and is independent of ρ_{vac} .

Proof. Define $S_{\mu\nu} \equiv G_{\mu\nu} - 8\pi G T_{\mu\nu}$. Taking the trace of (6) is trivially satisfied; however, the full information is contained in the traceless part. The tensor $S_{\mu\nu}$ has vanishing divergence by the Bianchi identity and energy conservation: $\nabla^\mu S_{\mu\nu} = 0$. Taking the trace of $S_{\mu\nu}$:

$$S \equiv g^{\mu\nu} S_{\mu\nu} = -R - 8\pi G T.$$

Equation (6) is equivalent to $S_{\mu\nu} - \frac{1}{4}S g_{\mu\nu} = 0$, which implies $S_{\mu\nu} = \frac{1}{4}S g_{\mu\nu}$. Since $\nabla^\mu S_{\mu\nu} = 0$, we have $\nabla_\nu S = 0$, so $S = -4\Lambda$ for some constant Λ . Then $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$, and Λ depends only on the initial data. $\square$

Counterarguments and Responses

Objection: Unimodular gravity still requires specifying Λ as a boundary condition. How does this help with the cosmological constant problem?

Response: The key shift is conceptual: in unimodular gravity, vacuum energy enters only through the traceless part of $T_{\mu\nu}$, and Λ is not sourced by ρ_{vac} . A constant shift in ρ_{vac} has no effect on the traceless equations. Therefore, the 120-order discrepancy between ρ_{vac} and Λ_{eff} ceases to be a problem *of the same type*: the question transforms from “why is there an extraordinary cancellation between Λ_{bare} and $8\pi G \rho_{\text{vac}}$?” into “why does the integration constant Λ take a small value?” [30, 31, 5].

This reformulation has been criticized on the grounds that it merely relocates the fine-tuning problem to initial or boundary conditions [5, 43]. Padmanabhan [5] has argued that the question “why is the boundary condition small?” is just as puzzling as the original cancellation, and Burgess [43] has emphasized that without a dynamical selection principle for the boundary data, unimodular gravity does not explain the observed value of Λ_{eff} .

These criticisms are legitimate but do not undermine the separation thesis as formulated here. The separation thesis does not claim to *solve* the cosmological constant problem in the sense of predicting Λ_{eff} from first principles. Rather, it claims that Λ and ρ_{vac} are *different quantities*—and that the apparent 120-order discrepancy arises from conflating them. In unimodular gravity, the question of why Λ is small is a question about *one* parameter (an integration constant), not about a miraculous cancellation between two enormous quantities. This is a genuine conceptual improvement, even if the deeper question of why the integration constant takes its observed value remains open [51, 49].

5.2. Codazzi equation and Cotton gravity formulations

Feng and Chen [12] have shown that gravitational theories constructed from the Codazzi equation—including Cotton gravity and gauge-gravity inspired models—also admit Λ as an integration constant. The Codazzi equation on a spacelike hypersurface Σ with extrinsic curvature K_{ab} and induced metric h_{ab} reads

$$D_b K_{ac} - D_c K_{ab} = R^d{}_{bca} n_d, \quad (7)$$

where D is the covariant derivative compatible with h_{ab} and n_d is the unit normal. Under certain algebraic closure conditions, integrating the constraint system yields the Einstein equations with Λ arising as a free constant.

5.3. Conformal Killing gravity

A more recent framework, conformal Killing gravity [32], also generates Λ as an integration constant without imposing the unimodular determinant condition $\sqrt{-g} = 1$. The field equations are obtained by requiring the Einstein tensor to satisfy a conformal Killing-type equation on the cotangent bundle; the cosmological constant then emerges from the integrability conditions.

5.4. Kaloper–Padilla vacuum energy sequestering

Kaloper and Padilla [52] have proposed an explicit dynamical mechanism—*vacuum energy sequestering*—that decouples radiative corrections to vacuum energy from the cosmological constant at the action level. Their construction introduces global (non-dynamical) variables into the gravitational action whose variational equations enforce a cancellation of the vacuum energy contribution to the spacetime curvature. The observed cosmological constant then arises as a residual integration constant determined by the entire spacetime history, not by local QFT vacuum loops.

This mechanism provides perhaps the most concrete realization of the separation thesis within a single action principle: vacuum energy is *dynamically sequestered* from Λ , regardless of the particle content. The construction is compatible with the Standard Model and does not require new light degrees of freedom. It should be noted that the mechanism relies on global variables whose physical interpretation remains debated [43], and the original formulation has been extended in several directions [53].

5.5. Degravitation

A related but distinct approach is *degravitation* [54, 55], in which vacuum energy gravitates weakly (or not at all) because the graviton has a small mass or the gravitational

coupling is frequency-dependent. In these models, long-wavelength sources—of which a constant ρ_{vac} is the extreme case—are filtered out by the gravitational response, so that only dynamical perturbations around the vacuum source curvature.

While degravitation models face theoretical challenges (ghost instabilities, strong-coupling issues), they illustrate a general point: the assumption that ρ_{vac} gravitates with the same strength as ordinary matter is a *physical hypothesis*, not a mathematical necessity. This is consistent with the conceptual claim in Remark 4.3.

5.6. Section Result and Implications

In at least three independent geometric reformulations of general relativity—trace-free gravity, Codazzi-equation theories, and conformal Killing gravity—the cosmological constant arises as an integration constant fixed by boundary data, with no dependence on quantum vacuum energy. The Kaloper–Padilla sequestering mechanism provides a *dynamical* realization of the same separation at the action level, while degravitation scenarios illustrate that the gravitational coupling of ρ_{vac} is a physical assumption rather than a logical necessity. Together, these approaches establish *geometric independence*: Λ_{geom} and ρ_{vac} are logically distinct, and multiple concrete mechanisms exist for their decoupling.

6. Mathematical Development II: Thermodynamic Sequestration

6.1. Volovik’s condensed-matter analogy

Volovik [15, 16] demonstrated that quantum liquids provide an exact analogy in which the cosmological constant problem is absent. In superfluid ^4He at zero temperature, the vacuum (ground-state) energy density has a microscopic estimate of order E_{Debye}^4 from zero-point phonon modes. However, the actual ground-state energy is not determined by this estimate; it is controlled by thermodynamic equilibrium.

In a quantum liquid at zero temperature and zero external pressure, the Gibbs–Duhem relation requires

$$\epsilon + P = \mu n, \quad (8)$$

where ϵ is the energy density, P the pressure, μ the chemical potential, and n the number density. At equilibrium with $P = 0$ and particle number fixed, minimizing the free energy yields

$$\epsilon_{\text{vac}} = \mu_{\text{eq}} n_{\text{eq}} - P_{\text{ext}}. \quad (9)$$

The microscopic zero-point contributions cancel exactly against trans-Planckian (atomic-scale) contributions, not by fine-tuning but by the thermodynamic stability of the

ground state.

Proposition 6.1 (Thermodynamic cancellation of zero-point energy). *In a stable quantum liquid at zero temperature with fixed particle number and vanishing external pressure, the vacuum energy density is determined by macroscopic parameters (equilibrium chemical potential and density) and is independent of the zero-point energies of the low-energy collective modes.*

Proof. The grand-canonical potential density at zero temperature is $\omega = \epsilon - \mu n$. Thermodynamic stability requires $\omega = -P$. At zero external pressure, $\omega = 0$, hence $\epsilon = \mu n$. This relation holds exactly for the total energy—including all microscopic (trans-Debye) contributions—independently of the zero-point sum over phonon modes. The phonon zero-point energy is thus absorbed into the microscopic equation of state and does not appear as a separate macroscopic observable. $\square$

6.2. Cosmological analogy

By analogy, Volovik argues that the cosmological Λ should be determined by macroscopic parameters of the universe—the Hubble scale H , Newton’s constant G , and the matter content—rather than by the zero-point energy of quantum fields [15]. The estimate

$$\Lambda_{\text{eff}} \sim H^2 \tag{10}$$

gives $\rho_\Lambda \sim 3H_0^2 M_{\text{Pl}}^2 \sim 10^{-47} \text{ GeV}^4$, consistent with observations and 120 orders of magnitude smaller than the QFT estimate.

Counterarguments and Responses

Objection: The condensed-matter analogy is suggestive but not rigorous: the universe is not a superfluid, and there is no known “microscopic” theory of quantum gravity that would enforce such cancellation.

Response: The analogy does not require the universe to be literally a superfluid. Its force lies in demonstrating that the zero-point energy of low-energy modes generically does *not* determine the ground-state energy in systems with an ultraviolet completion, because thermodynamic self-consistency provides an independent constraint. This undermines the implicit assumption of the Zel’dovich identification that zero-point sums are physically relevant for the gravitational sector [15, 17].

6.3. Section Result and Implications

Condensed-matter physics provides a concrete, soluble example in which zero-point energies do not determine the macroscopic “cosmological constant” of the emergent

theory. This supports the separation thesis by physical (thermodynamic) rather than purely mathematical means.

7. Mathematical Development III: Running Vacuum Models

7.1. Renormalization-group approach

Moreno-Pulido and Solà Peracaula [18, 19] have computed the renormalized vacuum energy density for quantum fields in FLRW spacetime using adiabatic regularization. Their central result is that ρ_{vac} evolves with the Hubble rate:

$$\rho_{\text{vac}}(H) = \rho_{\text{vac}}^{(0)} + \frac{3\nu}{8\pi} M_{\text{pl}}^2 H^2 + \mathcal{O}(H^4), \quad (11)$$

where ν is a small dimensionless coefficient determined by the particle content. For the Standard Model, $|\nu| \sim 10^{-3}$ [19]. Crucially, no quartic mass contributions $\sim m^4$ appear after proper renormalization, eliminating the need for fine-tuning.

Proposition 7.1 (Dynamical vacuum energy is inconsistent with a constant Λ). *If $\rho_{\text{vac}}(H)$ varies with H , and H evolves during cosmic expansion, then ρ_{vac} cannot be identified with a constant Λ in the Einstein equations.*

Proof. By definition, Λ in (1) is a spacetime constant: $\nabla_\mu \Lambda = 0$. However, (11) implies $d\rho_{\text{vac}}/dt = (\partial\rho_{\text{vac}}/\partial H)\dot{H} \neq 0$ during any epoch with $\dot{H} \neq 0$. Since the matter-dominated and dark-energy-dominated eras both have $\dot{H} \neq 0$, the running vacuum energy is time-dependent over cosmologically relevant timescales. Therefore $\rho_{\text{vac}}(H) \neq \Lambda/(8\pi G)$ during these epochs. $\square$

7.2. Connection to effective field theory on curved spacetime

The running (11) has a natural interpretation in terms of the Schwinger–DeWitt heat-kernel expansion for the one-loop effective action [28, 27, 26]. For a massive scalar field of mass m in a general curved background, the renormalized effective Lagrangian contains terms of the form

$$\mathcal{L}_{\text{eff}} \supset \frac{1}{(4\pi)^2} \left(\frac{m^2}{2} R + \frac{1}{6} R_{\mu\nu} R^{\mu\nu} - \frac{1}{180} R^2 + \dots \right), \quad (12)$$

where the ellipsis denotes higher-order curvature invariants. On an FRW background with $R = 6(2H^2 + \dot{H})$, these curvature terms translate into H -dependent contributions to the effective vacuum energy. The absence of quartic-mass terms in the renormalized result follows from the structure of the adiabatic subtraction scheme, which matches and removes the divergences order by order in the WKB expansion [29, 18].

Counterarguments and Responses

Objection: The running vacuum model modifies the standard Λ CDM predictions. Does it conflict with current data?

Response: The corrections are proportional to νH^2 with $|\nu| \sim 10^{-3}$, so deviations from Λ CDM are at the percent level. Current Bayesian analyses using CMB, BAO, and SNe Ia data find that the running vacuum model is competitive with—and in some tensions slightly preferred over—the standard model [19, 20]. The model is falsifiable by precision measurements of the growth rate $f\sigma_8(z)$ and the Hubble parameter $H(z)$.

Objection: Weinberg’s no-go theorem [3] demonstrates that no dynamical adjustment mechanism based on a scalar field can relax the cosmological constant to zero without fine-tuning. Does the running vacuum evade this theorem?

Response: Weinberg’s no-go theorem applies specifically to *adjustment mechanisms*—models in which a scalar field dynamically relaxes Λ_{eff} to zero via a feedback loop. The running vacuum model is not an adjustment mechanism in this sense: it does not predict that $\rho_{\text{vac}}(H)$ relaxes to zero but rather that it evolves as a power series in H^2 , with the zeroth-order term $\rho_{\text{vac}}^{(0)}$ remaining as a free parameter (analogous to the integration constant in unimodular gravity). The running vacuum therefore sidesteps rather than violates Weinberg’s theorem. Similarly, Brown’s QKDE framework [13] does not employ an adjustment mechanism: it parameterizes the kinetic normalization $K(\chi)$ as a given function, not as a dynamical field that adjusts Λ . The no-go theorem constrains a specific class of models; the approaches reviewed here belong to a different class.

7.3. Section Result and Implications

Renormalization-group methods show that the quantum vacuum energy density runs with H , making it dynamical and incompatible with a strictly constant Λ . This provides field-theoretic support for the separation thesis.

8. Mathematical Development IV: The QKDE Framework

8.1. Motivation

The approaches of Sections 5–7 establish that Λ and ρ_{vac} can be separated in principle. A natural question is whether an explicit *action principle* can be constructed in which:

- (a) the metric sector remains exactly Einstein–Hilbert (no modified gravity);
- (b) the scalar (dark-energy) sector carries all late-time phenomenology;
- (c) the Planck mass is constant and gravitational waves propagate luminally.

Brown’s Quantum-Kinetic Dark Energy (QKDE) framework [13] achieves precisely this. Published in 2026 as an open-access article in the *International Journal of Modern Physics D*, the QKDE paper develops a 71-page, self-contained construction—from ultraviolet operator basis through one-loop effective action, covariant completion via a clock (Stückelberg) field, background dynamics in e-fold variables, linear perturbation theory, EFT-DE mapping, forecast-ready observables, and a reproducible numerical pipeline with exact variational sensitivity equations. The mathematical formalism that Brown provides is, to the author’s knowledge, one of the most developed realizations of a GR-preserving dark-energy framework with a time-dependent kinetic normalization available in the current literature. As a recently published construction (2026), it has not yet been subjected to extensive independent replication or community scrutiny—a limitation acknowledged in Section 12. The present section summarizes the key elements of Brown’s construction and develops their implications for the separation thesis.

8.2. Brown’s QKDE action and its structure

The defining effective action, as formulated by Brown [13], is

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{pl}}^2 R + K(\chi) X - V(\phi) \right], \quad (13)$$

where $X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$ is the kinetic invariant of a scalar field ϕ , $V(\phi)$ is a scalar potential, and $K(\chi) > 0$ is a scalar function of a clock (Stückelberg) field χ whose background value defines cosmic time. Working in unitary gauge $\chi = t$, the action reduces to a background description with a time-dependent kinetic normalization $K(t) \equiv K(\chi)|_{\chi=t}$ [13].

Definition 8.1 (QKDE in EFT-DE language). In the Effective Field Theory of Dark Energy (EFT-DE) [21, 22], the QKDE framework corresponds to the Bellini–Sawicki parameter set

$$\alpha_K = \frac{K \dot{\phi}^2}{H^2 M_{\text{pl}}^2} > 0, \quad \alpha_B = \alpha_M = \alpha_T = 0. \quad (14)$$

The braiding is zero, the Planck mass is constant ($\alpha_M = 0$), and the tensor speed equals the speed of light ($\alpha_T = 0$, consistent with GW170817 [23, 24]).

8.3. Background equations

On a spatially flat FRW background $ds^2 = -dt^2 + a^2(t) d\mathbf{x}^2$, the stress-energy components of the scalar sector are

$$\rho_\phi = \frac{K \dot{\phi}^2}{2} + V(\phi), \quad p_\phi = \frac{K \dot{\phi}^2}{2} - V(\phi), \quad w_\phi = \frac{K \dot{\phi}^2/2 - V}{K \dot{\phi}^2/2 + V}. \quad (15)$$

Because the metric sector is pure GR, the Friedmann and Raychaudhuri equations retain their standard forms:

$$H^2 = \frac{1}{3M_{\text{pl}}^2}(\rho_m + \rho_r + \rho_\phi), \quad (16)$$

$$\dot{H} = -\frac{1}{2M_{\text{pl}}^2}\left(\rho_m + \frac{4}{3}\rho_r + K\dot{\phi}^2\right). \quad (17)$$

The scalar field obeys a modified Klein–Gordon equation that incorporates the time-dependent kinetic normalization:

$$K(t)(\ddot{\phi} + 3H\dot{\phi}) + \dot{K}(t)\dot{\phi} + V_{,\phi} = 0. \quad (18)$$

The additional friction term $\dot{K}\dot{\phi}$ arises from the covariant structure of the clock-field sector and represents energy exchange between the scalar and clock sectors [13].

8.4. Scalar sound speed and ghost freedom

Proposition 8.2 (Luminal propagation and ghost freedom in QKDE). *For $P(X, \phi, \chi) = K(\chi)X - V(\phi)$, the scalar sound speed is $c_s^2 = 1$ and ghost freedom requires $K(\chi) > 0$.*

Proof. The general scalar sound speed for a Lagrangian $P(X)$ is [25]

$$c_s^2 = \frac{P_{,X}}{P_{,X} + 2X P_{,XX}}. \quad (19)$$

For $P = K(\chi)X - V(\phi)$, we have $P_{,X} = K(\chi)$ and $P_{,XX} = 0$. Therefore $c_s^2 = K/K = 1$. Ghost freedom requires the kinetic coefficient to be positive, $P_{,X} = K(\chi) > 0$. $\square$

8.5. Linear perturbations and GR consistency

A central result of the QKDE framework is that linear perturbation theory remains exactly that of GR [13]. In Newtonian gauge,

$$ds^2 = -(1 + 2\Phi) dt^2 + a^2(t)(1 - 2\Psi) d\mathbf{x}^2, \quad (20)$$

the scalar sector contributes no anisotropic stress at linear order because $P_{,XX} = 0$. Therefore:

Theorem 8.3 (GR-like linear phenomenology of QKDE). *In the QKDE framework, at linear order and on subhorizon scales:*

(i) *The Bardeen potentials satisfy $\Phi = \Psi$.*

(ii) The Poisson equation takes the standard GR form:

$$-\frac{k^2}{a^2}\Phi = 4\pi G \rho_m \delta_m. \quad (21)$$

(iii) The phenomenological response functions are $\mu(a, k) = \Sigma(a, k) = 1$ and the gravitational slip vanishes: $\eta(a, k) \equiv \Phi/\Psi - 1 = 0$.

(iv) The linear growth factor $D(a)$ satisfies the standard GR equation:

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2}\Omega_m(a) D = 0, \quad (22)$$

where primes denote $d/d \ln a$.

Proof. Points (i)–(iii) follow from $\alpha_B = \alpha_M = \alpha_T = 0$ together with $c_s^2 = 1$ (Proposition 8.2). The vanishing braiding means the scalar sector does not kinetically mix with the metric perturbations. The Einstein–Hilbert metric sector then yields the standard GR constraints, and the dark-energy scalar perturbation $\delta\phi$ is pressure-supported on subhorizon scales (decaying as $(aH/k)^2$ relative to the matter perturbation [38, 39]), making it negligible in the Poisson equation. Point (iv) follows from substituting (21) into the matter continuity and Euler equations in the subhorizon limit [13, 37]. $\square$

8.6. Brown’s kinetic normalization specifications

Brown [13] analyzes two explicit forms for $K(t)$, each with a closed algebraic identity for K'/K that requires no iteration or recursion—a notable technical achievement that makes the framework immediately implementable in numerical pipelines.

Curvature-motivated form. Motivated by one-loop QFT in curved spacetime, where integrating out a heavy field of mass M generates dimension-six operators $\sim (R/M^2)X$ [27, 26, 28], one sets

$$K = 1 + \frac{\alpha R}{M^2}, \quad (23)$$

where α is a dimensionless Wilson coefficient. On FRW, $R = 6H^2(2 + H'/H)$, so K varies slowly with the expansion rate. An algebraic, iteration-free expression for K'/K is derived in [13], Eq. (4.10).

Phenomenological running. For model-space exploration, a simple exponential form is

$$K(N) = 1 + K_0 e^{pN}, \quad N = \ln a, \quad (24)$$

with $K'/K = p(K-1)/K$. Ghost freedom requires $K(N) > 0$ throughout the integration domain.

8.7. Structural analogy with running vacuum models

The curvature-motivated form (23) suggests a structural analogy—noted here as the present author’s interpretive observation, not as a claim of Brown’s original paper—with the running vacuum models of Section 7. In the QKDE framework, the scalar energy density is

$$\rho_\phi = \frac{K\dot{\phi}^2}{2} + V(\phi), \quad (25)$$

where $K = 1 + \alpha R/M^2$ and $R = 6(2H^2 + \dot{H})$. To leading order in $\alpha H^2/M^2$, the kinetic-normalization correction shifts the effective dark-energy density by

$$\delta\rho_\phi \sim \frac{\alpha R}{M^2} \cdot \frac{\dot{\phi}^2}{2} \sim \frac{\alpha}{M^2} \cdot H^2 \cdot (\text{kinetic energy}), \quad (26)$$

producing an H^2 -dependent contribution structurally analogous to the νH^2 term in the running vacuum model (11). It is important to emphasize that this is a *structural analogy*, not an identification: Brown’s original motivation for the curvature-motivated kinetic normalization concerns the dynamics of the kinetic sector (wave-function renormalization of a light scalar by heavy fields), not vacuum energy running directly [13, 14]. The two frameworks share an H^2 -dependent modification of the dark-energy sector but differ in their physical origin, their perturbative structure ($c_s^2 = 1$ identically in QKDE, while general k-essence permits $c_s^2 \neq 1$), and their relationship to the cosmological constant. The analogy is nevertheless suggestive: it indicates that the running vacuum phenomenology may admit a natural EFT-level embedding within a GR-preserving scalar-field construction.

Counterarguments and Responses

Objection: QKDE modifies the dark-energy sector but does not directly address whether vacuum energy gravitates. How does it support the separation thesis?

Response: Brown’s QKDE construction demonstrates that the full late-time dark-energy phenomenology—expansion history, growth of structure, distances—can be carried by a scalar sector with a running kinetic normalization, while the geometric sector (and hence Λ) remains strictly that of GR. There is no need to identify the “vacuum energy” of the scalar with Λ ; the two play distinct roles in Brown’s action (13). The explicit construction refutes any claim that late-time acceleration *requires* $\Lambda = 8\pi G \rho_{\text{vac}}$.

Objection: The QKDE framework is an effective field theory valid below some cutoff. Can it address the full cosmological constant problem, which involves Planck-scale

physics?

Response: No effective theory can resolve a problem at energy scales above its own cutoff. What Brown’s QKDE achieves is to show that, within the domain of validity of semiclassical gravity and the EFT-DE framework, there exists a consistent, falsifiable description in which Λ and ρ_{vac} are separated. The full resolution presumably requires a UV-complete theory of quantum gravity, but the EFT-level separation that Brown constructs is already physically meaningful and observationally testable.

8.8. Falsifiable predictions

Brown’s QKDE framework makes sharp, null predictions at linear order [13]:

$$\mu(a, k) = \Sigma(a, k) = 1, \quad \eta(a, k) = 0, \quad c_T^2 = 1. \quad (27)$$

Any statistically significant detection of $\mu \neq 1$, $\Sigma \neq 1$, nonzero gravitational slip, or subluminal/superluminal tensor propagation at late times would falsify this framework. Additionally, with $K > 0$ enforced, the null energy condition $\rho_\phi + p_\phi = K\dot{\phi}^2 \geq 0$ holds, so $w_\phi \geq -1$ and phantom behavior is excluded.

8.9. Section Result and Implications

Brown’s QKDE framework provides an explicit, mathematically rigorous action principle in which all dark-energy phenomenology is carried by a scalar sector with a time-dependent kinetic normalization, while the Einstein–Hilbert metric sector is unmodified. This demonstrates that the separation of Λ from ρ_{vac} can be realized at the level of an EFT with falsifiable predictions. The framework’s internal consistency—verified through exact diagnostic identities (Appendix B)—and its comprehensive treatment of stability, perturbation theory, and observational forecasts make it one of the most developed mathematical realizations of the separation thesis currently available, though it awaits independent replication and full confrontation with observational data.

9. Mathematical Development V: Conceptual and Foundational Analysis

9.1. The Rugh–Zinkernagel critique

Rugh and Zinkernagel [9, 10] have provided a systematic examination of the assumptions underlying the cosmological constant problem. Their analysis identifies three assumptions, each of which is contestable:

Assumption 1: Vacuum energy is gravitationally active. QFT computes ρ_{vac} as an energy density, but the physical reality of this quantity as a gravitational source is not directly established. The Casimir effect and Lamb shift are sometimes cited as evidence for “real” vacuum energy, but these effects can be computed without reference to zero-point energy (using source theory or the fluctuation–dissipation approach) [35, 36].

Assumption 2: The Standard Model is valid up to the Planck scale. The 120-order discrepancy assumes a cutoff at M_{Pl} ; with a TeV cutoff the mismatch reduces to ~ 55 orders. There is no empirical evidence that the Standard Model (or any known QFT) is valid at such scales.

Assumption 3: The bare cosmological constant has no independent physical role. If Λ_{bare} in (3) is treated as a free geometric parameter (as in unimodular gravity), then the “fine-tuning” is simply the statement that Λ_{bare} has been measured to have a specific (small) value, analogous to any other fundamental constant.

Remark 9.1 (Conceptual claim: Conditionality of the cosmological constant problem). The cosmological constant problem exists as a quantitative discrepancy if and only if all three of the following conditions hold: (a) ρ_{vac} gravitates in the same manner as ordinary matter; (b) the QFT computation of ρ_{vac} is valid up to some high cutoff $\Lambda_{\text{UV}} \gg H_0$; and (c) Λ_{bare} is not an independent geometric quantity. *Note:* This is a conditional philosophical claim, not a mathematical proposition; the argument below establishes the logical structure.

Argument. If (a) fails—i.e., ρ_{vac} does not source gravity or sources it differently—then the right-hand side of (3) does not contain ρ_{vac} , and no discrepancy arises. If (b) fails—i.e., the QFT estimate is unreliable above some moderate scale—then ρ_{vac} is not known to be large, and the discrepancy is at most of order $(\Lambda_{\text{UV}})^4/(8\pi G)$. If (c) fails—i.e., Λ_{bare} is a free constant as in unimodular gravity—then Λ_{eff} is simply the measured value of a free parameter, and the cancellation between Λ_{bare} and $8\pi G \rho_{\text{vac}}$ is not “fine-tuning” but determination of an initial condition. The problem requires all three conditions simultaneously. $\square$

Counterarguments and Responses

Objection: Questioning whether vacuum energy gravitates seems to undermine the entire semiclassical program.

Response: The semiclassical program is extraordinarily successful for processes where the gravitational backreaction of quantum fields can be tested—Hawking radiation, inflationary perturbations, particle creation in expanding universes. However, the

gravitational effect of a constant, homogeneous, infinite vacuum energy density has never been directly measured. What is tested is the *difference* in vacuum energy between configurations (e.g., Casimir plates), not the absolute value [35]. Questioning whether the absolute value gravitates is not a rejection of semiclassical gravity but a refinement of its domain of applicability.

9.2. Section Result and Implications

The cosmological constant problem rests on three separable assumptions, each of which can be questioned within established physics. This analysis provides conceptual support for the separation of Λ from ρ_{vac} without requiring any departure from known physics.

10. Empirical and Data-Grounded Illustrations

10.1. Observational status of Λ CDM

The concordance Λ CDM model, calibrated by Planck CMB data [40], provides $\Omega_\Lambda = 0.6847 \pm 0.0073$ and $H_0 = 67.36 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$. These values imply

$$\rho_\Lambda = \frac{\Lambda_{\text{eff}}}{8\pi G} = 3\Omega_\Lambda H_0^2 M_{\text{pl}}^2 \approx 3.6 \times 10^{-47} \text{ GeV}^4. \quad (28)$$

This is the quantity that the QFT estimate of ρ_{vac} must match if the Zel'dovich identification holds.

10.2. Running vacuum predictions and current data

The running vacuum model (11) with $|\nu| \sim 10^{-3}$ predicts mild deviations from Λ CDM in the growth rate $f\sigma_8(z)$. Solà Peracaula and collaborators [19, 20] have reported that combined analyses of CMB+BAO+SNe Ia+ $H(z)$ data mildly favor $\nu > 0$ at the 2–3 σ level, though the result is not yet definitive.

It should be noted that these results predate the DESI 2024 BAO data release, which has provided new constraints on the dark-energy equation of state. The DESI results [56] suggest mild evidence for time-varying dark energy ($w_0 > -1$, $w_a < 0$ in the CPL parameterization), which is *qualitatively consistent* with the running vacuum prediction of a mildly dynamical vacuum equation of state [18]. However, a dedicated analysis of the running vacuum model against DESI data has not yet been published, and such an analysis is needed before drawing firm conclusions. If confirmed by future DESI data releases and by Euclid, this would provide direct evidence for the dynamical nature of vacuum energy and against the identification $\rho_{\text{vac}} = \text{const}$.

10.3. QKDE observational signatures and projected constraints

Within QKDE, the null predictions (27) can be tested by:

- Gravitational lensing surveys (Euclid, Rubin/LSST) measuring $\Sigma(a, k)$;
- Redshift-space distortion measurements of $f\sigma_8(z)$ constraining $\mu(a, k)$;
- Gravitational-wave standard sirens constraining c_T and the Planck mass evolution.

The curvature-motivated kinetic normalization (23) introduces deviations from Λ CDM only through the background expansion history $H(a)$, preserving the GR growth equation (22). Brown [13], Sec. 9, develops a Fisher-matrix forecast based on exact variational (sensitivity) equations for distances, $H(z)$, and $f\sigma_8(z)$. The sensitivity system co-evolves parameter derivatives alongside the background, providing exact (non-finite-differenced) Jacobians for the Fisher assembly. While the numerical results depend on the assumed survey specifications and fiducial parameters, the pipeline itself is fully specified and reproducible—a quantitative forecast awaits implementation with specific survey configurations (DESI, Euclid, Rubin/LSST).

10.4. Section Result and Implications

Current observational data are consistent with Λ CDM but do not exclude mild running of the vacuum energy or a QKDE-type kinetic modification at the background level. Forthcoming surveys will provide decisive tests.

11. Synthesis and Discussion

The five lines of evidence developed in Sections 5–9 converge on a common conclusion that can be stated precisely:

Theorem 11.1 (Separation thesis — synthesis of existing results). *There exist mathematically consistent and observationally viable formulations of gravitational theory in which:*

- (i) *The cosmological constant Λ is a geometric quantity whose value is independent of the quantum vacuum energy density ρ_{vac} .*
- (ii) *The quantum vacuum energy density ρ_{vac} evolves dynamically with the Hubble rate and is not a spacetime constant.*
- (iii) *All late-time dark-energy phenomenology can be carried by a scalar sector without modifying the Einstein–Hilbert metric sector.*

Therefore, the identification $\Lambda = 8\pi G \rho_{\text{vac}}$ is not a necessary consequence of established physics, and the cosmological constant problem may be an artifact of this identification rather than a crisis requiring exotic new physics.

Remark on novelty. *The individual components of this theorem are not new: unimodular gravity dates to Anderson and Finkelstein [50], the running vacuum program to Solà Peracaula [11], and the foundational critique to Rugh and Zinkernagel [9]. The novelty lies in the synthesis—assembling these independent lines into a unified logical structure—and in the integration of Brown’s QKDE framework [13], which provides Statement (iii) with an explicit action principle not previously available.*

Proof. Statement (i) is established by Propositions 4.3 and 5.1 (geometric approaches). Statement (ii) is established by Proposition 7.1 (running vacuum models). Statement (iii) is established by Theorem 8.3 and Proposition 8.2, which follow from Brown’s QKDE framework [13]. The three statements are logically independent: each is derived from a different mathematical structure and relies on different physical assumptions. Statement (iii) in particular depends on Brown’s construction of a covariantly completed action with $\alpha_B = \alpha_M = \alpha_T = 0$, which is the only known explicit realization of the separation at the EFT level with a complete perturbation theory and numerical pipeline. $\square$

Counterarguments and Responses

Objection: The separation thesis does not *solve* the cosmological constant problem; it merely *dissolves* it by denying the premises.

Response: Precisely. If the premises are not well-founded—and each of the three assumptions in Proposition 9.1 can be independently questioned—then the problem is indeed dissolved rather than solved. This is not an evasion but a conceptual clarification, analogous to the dissolution of the luminiferous-aether problem upon recognizing that the aether hypothesis was not required by electromagnetic theory.

Objection: Even if Λ and ρ_{vac} are formally distinct, quantum gravity presumably relates them. The separation may be an artifact of our ignorance.

Response: This objection deserves serious engagement. The separation thesis operates within the regime of semiclassical gravity and effective field theory, which is the regime in which the cosmological constant problem is formulated. Crucially, however, Volovik’s condensed-matter analogy (Proposition 6.1) provides a concrete counterexample from a system with a *known* UV completion: in superfluid helium, the full microscopic theory (nonrelativistic many-body quantum mechanics) is completely understood, and it *does not* restore the identification between zero-point energy and the macroscopic “cosmological constant” of the emergent low-energy theory [15, 16]. The UV completion instead enforces thermodynamic self-consistency, which decouples the two quantities even more sharply.

This does not prove that quantum gravity will behave similarly, but it establishes that the relationship between Λ and ρ_{vac} in a UV-complete theory can be *very different*

from the Zel’dovich identification—and that the separation can be *strengthened*, not weakened, by understanding the underlying microphysics. If a UV-complete theory of quantum gravity ultimately links Λ and ρ_{vac} , the nature of that link may resemble thermodynamic equilibrium more than naïve energy counting.

12. Limitations

This review is subject to several limitations:

First, the geometric approaches (Section 5) establish that Λ *can* be an integration constant, but they do not prove that it *must* be. The standard formulation in which Λ is sourced by matter remains a viable interpretation.

Second, the condensed-matter analogy (Section 6) is suggestive but not a proof. The universe may lack the thermodynamic features (equilibrium, fixed particle number) that enforce the cancellation in superfluid helium.

Third, the running vacuum model (Section 7) and the QKDE framework (Section 8) are effective descriptions valid below some UV cutoff. They do not address the full cosmological constant problem at Planck-scale energies.

Fourth, Brown’s QKDE framework [13] was published in 2026 and has not yet been subjected to independent replication, full Bayesian data analysis using current cosmological data (CMB, BAO, SNe Ia, RSD), or extensive community scrutiny. The Fisher-matrix setup provided by Brown in [13], Sec. 9, furnishes the necessary forecast-ready machinery, but the analysis remains to be carried out. Until independent groups have reproduced and tested the framework, the results attributed to QKDE in this review should be regarded as established within the internal logic of the construction but awaiting external validation.

Fifth, all results in this review pertain to the linear regime. Nonlinear clustering, baryonic feedback, and screening mechanisms are beyond the scope of the present analysis.

13. Future Directions

- (1) **Data confrontation.** Apply the Fisher and Bayesian pipelines developed by Brown [13], Sec. 9, to DESI, Euclid, and Rubin/LSST data to constrain the QKDE parameters (α, M) or (K_0, p) . Brown’s exact variational (sensitivity) equations for distances and $f\sigma_8(z)$ provide the necessary forecast-ready machinery.
- (2) **Boltzmann-solver integration.** Couple Brown’s QKDE background system to a Boltzmann code (CLASS or CAMB) to compute CMB power spectra and test for early-time kinetic running.

- (3) **Nonlinear regime.** Map QKDE background histories, as generated by Brown’s numerical pipeline (Sec. 8 and Appendix I of [13]), onto halo-model emulators to predict nonlinear lensing and clustering statistics.
- (4) **UV completion.** Investigate whether a specific UV-complete theory (e.g., asymptotic safety, string theory) predicts the separation of Λ from ρ_{vac} and, if so, with what quantitative relation.
- (5) **Cross-disciplinary connections.** Explore whether the separation thesis has analogs in other areas of physics where emergent constants are distinct from microscopic parameters—information geometry, complex systems, and condensed-matter physics may provide structural insights.

14. Position Within the Research Program

This article is Paper **#1a** in the comprehensive research program “*What If the Vacuum Gravitates Locally? Separating Cosmic Expansion from Quantum Vacuum Energy*,” developed at the Institute of Integrative and Interdisciplinary Research (Toronto) and the Information Physics Institute (Gosport). The program comprises 17 papers organized into seven parts, each exploring a different aspect of the separation thesis from established physics—general relativity, quantum field theory, the Standard Model, and $E = mc^2$ —without invoking new particles, new forces, or fine-tuned constants.

The present paper provides the *mathematical and conceptual foundation* for the entire program: it establishes rigorously that the identification $\Lambda = 8\pi G \rho_{\text{vac}}$ is not a theorem of known physics, and it assembles the mathematical machinery—from geometric formulations and thermodynamic analogies to Brown’s QKDE effective-field-theory framework [13]—on which subsequent papers build.

A note on transparency. The reader may wonder whether listing the full program in a review article risks blurring the boundary between a neutral scholarly synthesis and a programmatic manifesto. The author has chosen transparency over concealment: it is better for the reader to know explicitly that this paper serves a dual role—as an independent review of established literature *and* as the theoretical anchor of a larger research program—than to discover this context indirectly. Every proposition, theorem, and derivation in Sections 5–9 stands or falls on its own mathematical merits, independently of the program. The program listing is provided so that readers can assess the author’s perspective and potential biases; it does not affect the validity of the arguments.

14.1. Complete list of papers in the program

The full program is listed below. Papers marked with ■ are complete (PDF available); those marked with □ are in draft or outline stage. The present paper is highlighted.

Part I: Foundations — The Central Hypothesis

- 1. *On Quantum Vacuum Energy, Cosmological Constant and Missing Mass — Asking the Right Questions.*
- 1a. **Theoretical Separation of Quantum Vacuum Energy from the Cosmological Constant: A Review of Foundational Approaches [this paper].**
- 2. *What If the Vacuum Gravitates Locally? — Separating Cosmic Expansion from Quantum Vacuum Energy.*

Part II: Theoretical Microphysics — Deriving the Vacuum–Matter Coupling

- 3. *A Microscopic Model for the Dependence of Vacuum Energy on Matter Density — Towards a Quantum Field Theory Derivation.*
- 4. *Matter-Dependent Vacuum Energy Density and Inhomogeneous Cosmic Expansion — A Phenomenological Approach.*
- 5. *A Sequestering Mechanism for Vacuum Energy — Why the Vacuum Expands Space Globally but Gravitates Locally.*

Part III: Cosmological Consequences — From Inflation to Large-Scale Structure

- 6. *Connection of Vacuum Energy with Inflation — Initial Conditions and the Origin of Primordial Fluctuations.*
- 7. *Structure Growth in the Gravitating Vacuum Model — Modified Perturbation Equations and the S_8 Tension.*
- 8. *N-body Simulations with a Gravitating Vacuum Phase — Testing the Model Against Structure Formation.*

Part IV: Galactic and Astrophysical Manifestations — From First Stars to Rotation Curves

- 9. *Known Properties of Vacuum Energy, Dark Matter and JWST Early Galaxy Formation — A Unified View.*
- 10. *Vacuum Energy and the Cosmic Microwave Background — Compatibility with Observed CMB Anisotropies.*

- 11. *Two Pathways of Primordial Cloud Collapse — Fragmentation versus Direct Collapse under Enhanced Vacuum Energy.*
- 12. *The Vacuum Capture Model — Phase Transitions and Galactic Rotation without Dark Matter.*
- 13. *The Gravity of Emptiness — Cosmic Voids as Anchors of the Filamentary Network.*

Part V: The Hidden Baryonic Sector — Compact Remnants and Cold Gas

- 14. *Invisible Gas as a Major Component of the Missing Mass — Fractal Cold Gas in Galactic Halos.*
- 15. *Mass of Compact Remnant Population in the Milky Way — Accounting for Direct Collapse Black Holes over 13 Billion Years.*

Part VI: Precision Tests — Supernovae and the Hubble Tension

- 16. *Type Ia Supernovae in a High-Density Vacuum Cosmology — Revisiting the Chandrasekhar Limit and Distance Calibration.*

Part VII: Philosophical Synthesis — The Copernican Conclusion

- 17. *The Copernican Principle and the End of Anthropocentrism in Cosmology — Why the Vacuum Gravitating Is More Elegant than Exotic Particles.*

14.2. Role of the present paper

Paper #1a occupies a distinctive position in the program. While Paper #1 poses the foundational questions historically and Paper #2 develops the core hypothesis physically, the present paper provides the *rigorous mathematical and theoretical review* that connects the separation thesis to the broader literature. Specifically:

- It establishes, through Propositions 4.3–9.1 and Theorem 11.1, that the identification $\Lambda = 8\pi G \rho_{\text{vac}}$ is a hypothesis rather than a theorem—providing the logical foundation on which all subsequent papers rest.
- It integrates Brown’s QKDE framework [13] into the program, supplying the EFT-level action principle (Section 8) that Papers #3, #5, and #12 will use for microphysical derivations and galactic modeling. Brown’s curvature-motivated kinetic normalization $K = 1 + \alpha R/M^2$ is particularly relevant to Paper #12 (The Vacuum Capture Model), where the Ricci scalar R differs between bound overdense regions and the homogeneous background—providing a natural mechanism for environment-dependent vacuum behavior.

- It synthesizes geometric (unimodular gravity), thermodynamic (Volovik), field-theoretic (running vacuum), and conceptual (Rugh–Zinkernagel) approaches into a unified argument, providing the interdisciplinary framework that the program’s later papers inherit.
- It identifies the falsifiable null predictions (Eq. (27)) that Papers #7, #8, and #16 will test against CMB, structure growth, and supernova data.

All preprints in the program are available on request from the author. Papers in PDF form are also accessible through the program page at the Institute of Integrative and Interdisciplinary Research.¹

14.3. Section Result and Implications

The present paper serves as the mathematical and theoretical anchor of a 17-paper research program. By establishing the logical independence of Λ and ρ_{vac} and by integrating Brown’s QKDE framework as the program’s EFT-level foundation, it provides the rigorous groundwork on which subsequent papers—from microphysical derivations to observational tests—are built.

15. Recommended Journal for Publication

This article is recommended for submission to **Foundations of Physics** (Springer).

The recommendation is based on the following considerations. The manuscript addresses a foundational question—the logical and physical relationship between the cosmological constant and quantum vacuum energy—that sits at the intersection of general relativity, quantum field theory, and philosophy of physics. *Foundations of Physics* has a long history of publishing work that critically examines the conceptual underpinnings of physical theories, including contributions by Volovik, Rugh, Zinkernagel, and others cited here. The journal’s readership spans the disciplines from which this review draws (cosmology, high-energy theory, condensed-matter physics, philosophy of science), making it an ideal venue for an interdisciplinary synthesis.

The mathematical content (propositions, theorems, EFT-DE mapping drawing extensively on Brown’s QKDE framework [13]) is rigorous but accessible, consistent with the journal’s standard of combining formal development with conceptual clarity. The falsifiable predictions (Section 8, Eq. (27)) ensure that the paper is not merely philosophical but connects to ongoing observational programs.

Alternative venues that could be considered include *Classical and Quantum Gravity* (IOP), *European Physical Journal C* (Springer), and *Studies in History and Philosophy of Modern Physics* (Elsevier), depending on the emphasis desired.

¹<https://interdisciplinary-research.institute/2026/03/11/research-program-invitation/>

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Appendices

A. Derivation of the QKDE E-fold System (after Brown [13])

For reference, we reproduce the closed autonomous system developed by Brown [13] in e-fold time $N = \ln a$. This system, presented in Sections 3–4 of Brown’s paper, is one of the key technical contributions of the QKDE framework: it reduces the full background cosmology to a first-order autonomous system that is both analytically transparent and numerically stable. Define the state vector $y = (\phi, s, y_H) \equiv (\phi, \phi', \ln H)$. Using the chain rules $\dot{f} = H f'$ and $\ddot{f} = H^2 f'' + H H' f'$, the background system (16)–(18) becomes:

$$\phi' = s, \quad (29)$$

$$s' = - \left(3 + \frac{H'}{H} + \frac{K'}{K} \right) s - \frac{V_{,\phi}}{H^2 K}, \quad (30)$$

$$y'_H = \frac{H'}{H} = - \frac{1}{2M_{\text{pl}}^2 H^2} (\rho_m + \frac{4}{3}\rho_r + K H^2 s^2), \quad (31)$$

with $\rho_m(N) = \rho_{m,0} e^{-3N}$, $\rho_r(N) = \rho_{r,0} e^{-4N}$, and $H = e^{y_H}$.

The kinetic invariant and scalar energy density are

$$X = \frac{1}{2} H^2 s^2, \quad \rho_\phi = \frac{1}{2} K H^2 s^2 + V(\phi). \quad (32)$$

For the curvature-motivated specification $K = 1 + \alpha R/M^2$, the key identity is

$$\frac{K'}{K} = \frac{c A}{1 - c B K}, \quad (33)$$

where $c = (\alpha/M^2) H^2 / (1 + \alpha R/M^2)$, $B = 3s^2/M_{\text{pl}}^2$, and

$$A = 24E + \frac{18K s^2}{M_{\text{pl}}^2} + \frac{9\rho_m + 16\rho_r + 6sV_{,\phi}}{M_{\text{pl}}^2 H^2}, \quad (34)$$

with $E = H'/H$. This expression is algebraic and iteration-free.

B. Diagnostic Identities for Numerical Validation (after Brown [13])

Brown [13], Section 8.5, specifies the following residuals, which vanish analytically and should remain at machine precision ($\lesssim 10^{-9}$) during numerical integration. These diagnostics constitute a rigorous self-consistency check that distinguishes Brown’s framework from less carefully validated constructions in the literature:

- (i) **Friedmann closure:** $C_F = 1 - \Omega_m - \Omega_r - \Omega_\phi$.

- (ii) **Raychaudhuri residual:** $C_R = y'_H + \frac{1}{2M_{\text{pl}}^2 e^{2y_H}} (\rho_m + \frac{4}{3}\rho_r + K e^{2y_H} s^2)$.
- (iii) **Klein–Gordon residual:** $C_\phi = s' + (3 + E + K'/K) s + V_{,\phi}/(H^2 K)$.
- (iv) **Ricci identity:** $C_{R/H^2} = R/e^{2y_H} - 6(2 + y'_H)$.

These diagnostics provide a rigorous self-consistency check for any numerical implementation.

Peer Reviews and Revision History

First Round

[Placeholder for first round of peer review and author responses.]

Second Round

[Placeholder for second round of review.]

Further Clarifications

[Placeholder for additional clarifications requested during review.]

Extended Formalization

[Placeholder for any extended formal developments added in response to referee comments.]